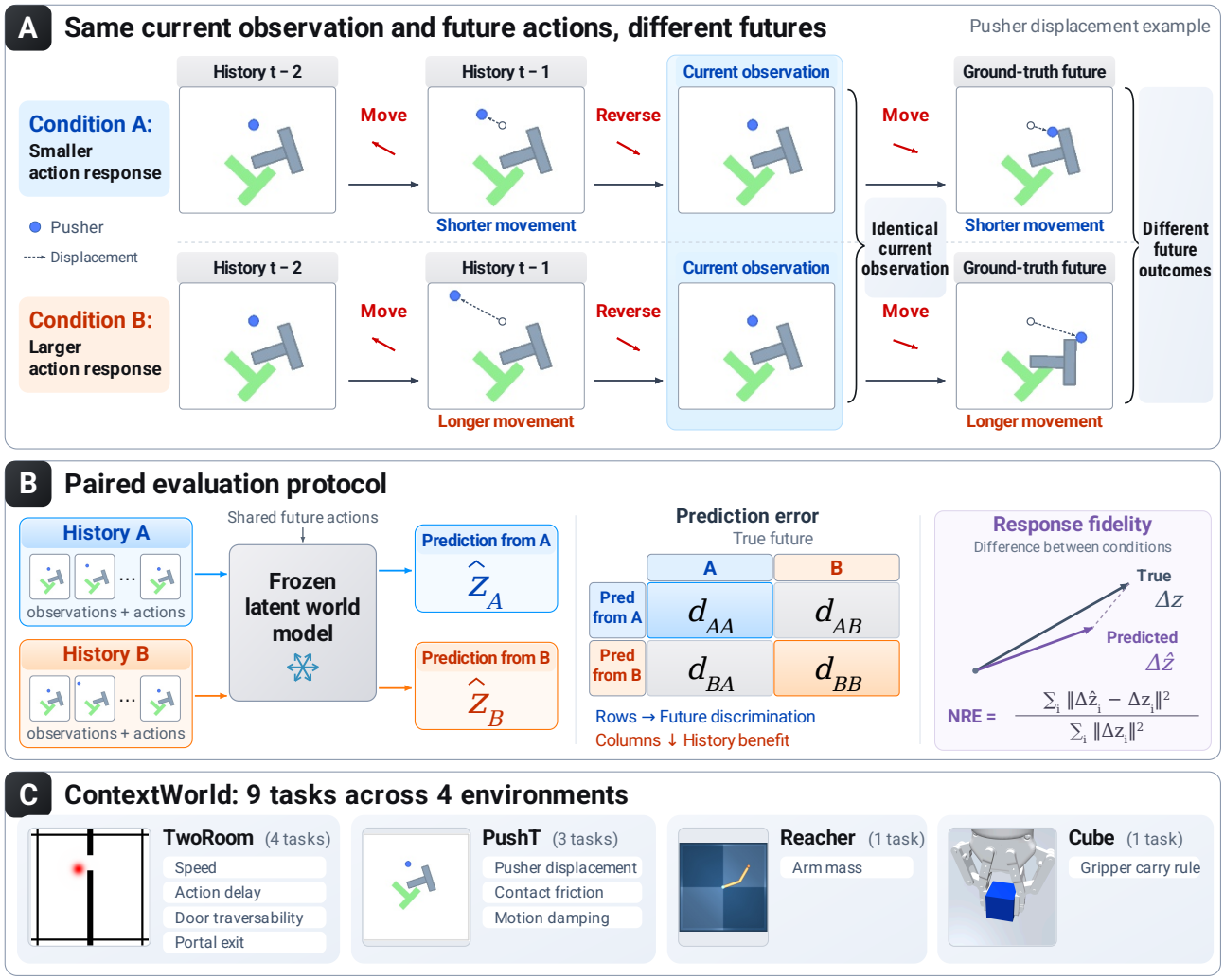
**Overview with response fidelity**




**ContextWorld overview.** (A) A recorded pusher-displacement pair. Shared actions reveal different movement distances in the history, return both conditions to the same current observation, and produce different future states. Red arrows summarize action directions; dashed arrows show observed endpoint displacement. All images use the same crop of the original observations. (B) A frozen model predicts a future from each history and the shared future actions. In the error matrix, $d_{AB}$ compares the prediction from history A with true future B. Row comparisons measure future discrimination; column comparisons measure the benefit of matching history for a fixed target. Response fidelity compares the predicted difference $\Delta\hat{z} = \hat{z}^B - \hat{z}^A$ with the true difference $\Delta z = z^B - z^A$. NRE measures their discrepancy, with zero indicating an exact response. The arrows illustrate this comparison schematically; they are not a projection of measured model outputs. (C) Nine tasks across four environments.